Assignment 10 :

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Code explanation:

a. Define Images of three:

Here , the three images named in my code image first , image second , image third are being defined using the array by NumPy library . Basically, these images are a second order 2*2 matrix which is represented using the pixel values.

b. Defining the coordinates:

Here , two sets of coordinates have been defined named 'horizontal coordinates newer ' and the 'vertical coordinates new ' are being defined to detect the horizontal and the vertical end points of the image.

c. Definition of the function which does the work :

The function 'apply coordinates_to_image_new' is being defined in order to calculate the features of an image which is based on the coordinates provided . What it does is basically to perform the element wise multiplication and the summation .

d. Visualization of the image :

Here , the image is then visualized using the pyplot library and the imshow() which is available from matlob . the role is display the image with the pixel values as colors.

e. Calculation the feature and display:

Here ,the function 'apply_coordinates_to_image' is being called twice which is done to calculate the horizontal and the vertical end points of the features scores and then printed.

So, this code overall code defines how to define the images , and then specify the coordinates for the feature selection and then visualize the images by calculating the features which are based on coordinates.

The execution code :

```
import numpy as num
from matplotlib import pyplot as pltt

# New images
imageee_first = num.array([num.array([120, 120]), num.array([90, 90])])
imageee_second = num.array([num.array([120, 120]), num.array([60, 0])])
imageee_third = num.array([num.array([120, 60]), num.array([120, 0])])

# New cooredinates for detecting horizontal and vertical end points
horizontal_coordinates_new = num.array([num.array([4, 4]), num.array([-4, -
4])])
vertical_coordinates_new = num.array([num.array([4, -4]), num.array([4, -4])])

# Function to apply the set points  to image
def apply_the_coordinates_to_image_new(image_new, coordinates_new):
   return num.sum(num.multiply(image_new, coordinates_new))

# plotting the image .
pltt.imshow(imageee_first)
pltt.axis('off')
pltt.title('Image of the First')
pltt.show()

# performing a check for the horiz and vert features applied to first image .
print('--- Imagee 1 ---')
print('Horizontalss extemity pointss featurees scoress:',
apply_the_coordinates_to_image_new(imageee_first,
horizontal_coordinates_new))
print('Verticals extremity points features scoress:',
apply_the_coordinates_to_image_new(imageee_first,
vertical_coordinates_new))
print()
```

```
# reprsenting the  image 2
pltt.imshow(imageee_second)
pltt.axis('off')
pltt.title('Image_of the  Second')
pltt.show()

# Checking for horizontal and vertical features in image 2
print('--- Imageee 2 ---')
print('horizontale end points features scores:',
apply_the_coordinates_to_image_new(imageee_second,
horizontal_coordinates_new))
print('down end points features score:',
apply_the_coordinates_to_image_new(imageee_second,
vertical_coordinates_new))
print()

# Visualizing image 3
pltt.imshow(imageee_third)
pltt.axis('off')
pltt.title('Image Third')
pltt.show()

# New coordinates for image 3
horizontal_coordinates_new_image3 = num.array([num.array([3, 3]),
num.array([-3, -3])])
vertical_coordinates_new_image3 = num.array([num.array([3, -3]),
num.array([3, -3])])

# Checking for accros and the down features present in the image 3rd
print('--- Image 3 ---')
print('across end points features score:',
apply_the_coordinates_to_image_new(imageee_third,
horizontal_coordinates_new_image3))
print('down end points features score:',
apply_the_coordinates_to_image_new(imageee_third,
vertical_coordinates_new_image3))
```

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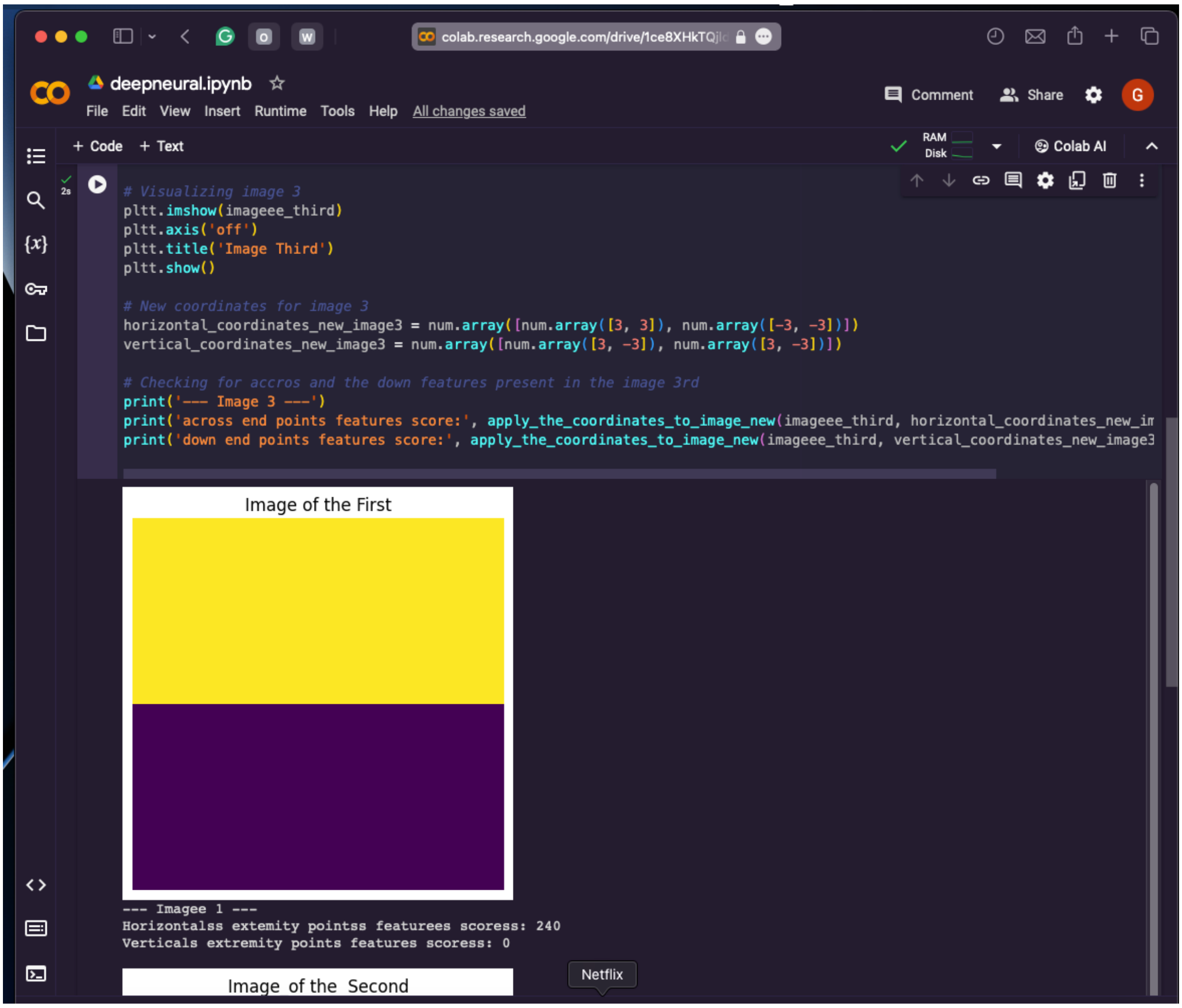
```python
import numpy as num
from matplotlib import pyplot as pltt

# New images
imageee_first = num.array([num.array([120, 120]), num.array([90, 90])])
imageee_second = num.array([num.array([120, 120]), num.array([60, 0])])
imageee_third = num.array([num.array([120, 60]), num.array([120, 0])])

# New cooredinates for detecting horizontal and vertical end points
horizontal_coordinates_new = num.array([num.array([4, 4]), num.array([-4, -4])])
vertical_coordinates_new = num.array([num.array([4, -4]), num.array([4, -4])])

# Function to apply the set points  to image
def apply_the_coordinates_to_image_new(image_new, coordinates_new):
    return num.sum(num.multiply(image_new, coordinates_new))

# plotting the image .
pltt.imshow(imageee_first)
pltt.axis('off')
pltt.title('Image of the First')
pltt.show()

# performing a check for the horiz and vert features applied to first image .
print('--- Imagee 1 ---')
print('Horizontalss extemity pointss featurees scoress:', apply_the_coordinates_to_image_new(imageee_first, horizontal_coo
print('Verticals extremity points features scoress:', apply_the_coordinates_to_image_new(imageee_first, vertical_coordinat
print()

# reprsenting the  image 2
pltt.imshow(imageee_second)
pltt.axis('off')
pltt.title('Image_of the  Second')
pltt.show()

# Checking for horizontal and vertical features in image 2
print('--- Imageee 2 ---')
print('horizontale end points features scores:', apply_the_coordinates_to_image_new(imageee_second, horizontal_coordinates
print('down  end points features score:', apply_the_coordinates_to_image_new(imageee_second, vertical_coordinates_new))
print()

# Visualizing image 3
pltt.imshow(imageee third)
```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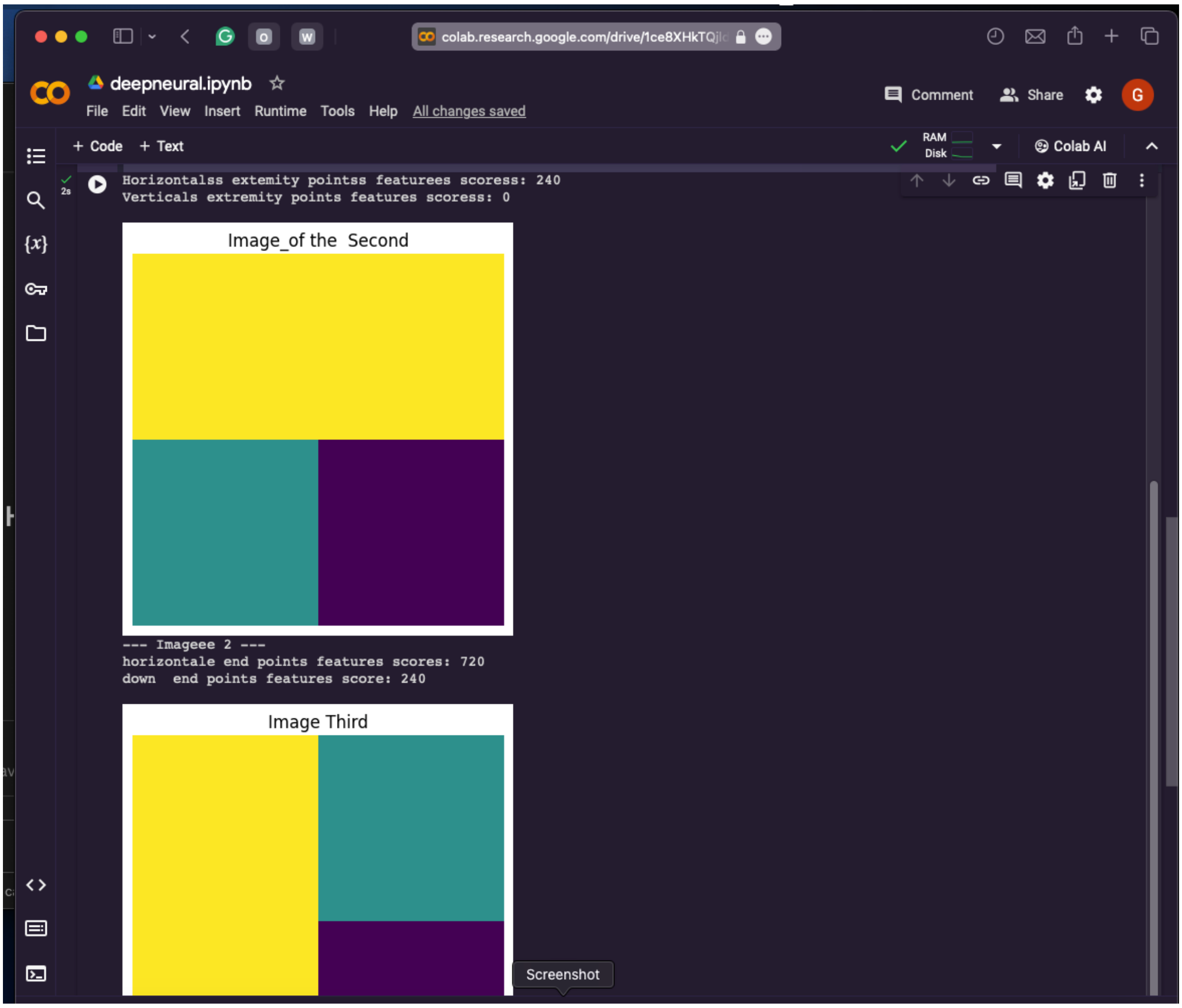
# Visualizing image 3
pltt.imshow(imageee_third)
pltt.axis('off')
pltt.title('Image Third')
pltt.show()
# New coordinates for image 3
horizontal_coordinates_new_image3 = num.array([num.array([3, 3]), num.array([-3, -3])])
vertical_coordinates_new_image3 = num.array([num.array([3, -3]), num.array([3, -3])])
# Checking for accros and the down features present in the image 3rd
print('--- Image 3 ---')
print('across end points features score:', apply_the_coordinates_to_image_new(imageee_third, horizontal_coordinates_new_im
print('down end points features score:', apply_the_coordinates_to_image_new(imageee_third, vertical_coordinates_new_image3
Image of the First
--- Imagee 1 ---
Horizontalss extemity pointss featurees scoress: 240
Verticals extremity points features scoress: 0
Image of the Second
Netflix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Horizontalss extemity pointss featurees scoress: 240
Verticals extremity points features scoress: 0
Image_of the Second
--- Imageee 2 ---
horizontale end points features scores: 720
down end points features score: 240
Image Third
Screenshot